

## PROJECT DESCRIPTION

# RarePath Evidence Atlas

**An evidence-first research atlas for rare childhood diseases**

## THE PROBLEM

Families and patient groups face fragmented evidence across reports, gene and variant records, publications, trials and research assets. A shared symptom can resemble another disease without establishing a shared molecular mechanism. Conventional searches rarely turn this complexity into a transparent next action.

## THE SOLUTION

RarePath creates a longitudinal case profile with structured symptoms, genomics, history, medications, family observations and biological evidence. It searches public biomedical sources, then maps disease → evidence → mechanism → possible connection → existing asset → collaborator → therapeutic research → validation gap → shared action.

## HOW OPENAI IS USED

OpenAI extracts claims from retrieved records, reconciles names and identifiers, helps plan the next research question and explains the evidence path. Each retained claim must point back to a source. The model is not itself evidence; uncited relationships are not promoted to graph edges.

## SAFETY BY DESIGN

Disease/mechanism evidence, therapeutic research evidence and relevance to this particular case are assessed separately. Low case applicability does not hide disease-level research, but it reduces the confidence of applying it to the child. A supported cluster is never automatically supported for this patient. Trial status is not efficacy, and a research lead is not a treatment recommendation.

# A decision, not a cure claim.

## Two scientifically honest endings

### A / JUSTIFIED COLLABORATION

Only when the cross-disease connection, reusable asset, named partner and shared next action have adequate source support. The path names what transfers, what does not, and what still needs validation.

### B / HONEST GAP + PLAN

When the bridge is not established, RarePath keeps mechanisms separate, identifies what is known and missing, shows the sources searched, and proposes the next question and investigation steps.

### WHAT CAN BE DEMONSTRATED

Quick and detailed case entry; explicit confirmation of report findings and HPO suggestions; evidence graph with citations; therapeutic research landscape; reusable research assets; people and partners from source records; mechanism clusters; next-question conversation; and three guided demos. The Rett/MECP2 versus GLUT1/SLC2A1 example ends with an honest gap, not an unsupported merged mechanism.

### CURRENT LIMITATIONS

This is a research prototype, not a diagnostic system, clinical trial matcher or prescribing tool. Live source availability affects results. Case profiles and analyses are saved in this browser on this device, without accounts or cloud sync. The guided demo does not yet use a fixed evidence snapshot. The default sample cases are illustrative; they are not real patients.

### DEMO & CONTACT

**Research walkthrough:** <https://youtu.be/xQQPJHlnshE>

**Product demo:** <https://youtu.be/ksX-CVb7DUA>

**Team introduction:** <https://youtu.be/r1tjXj0CasM>

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